EXP NO – 08 : BANKERS DEADLOCK AVOIDANCE ALGORITHM

```c
#include<stdio.h>
int main(){
        int n,r,i,j,k;
        printf("Enter number of processes");
        scanf("%d",&n);
        printf("Enter number of resources types");
                scanf("%d",&r);
        int alloc[n][r],max[n][r],need[n][r],avail[r];
        int finish[n],safe_sequence[n];
        int count=0,safeFound;
        printf("Enter Aloocation Matrix:\n");
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&alloc[i][j]);
                }
        }
        printf("Enter Max Matrix:\n");
        for(i=0;i<n;i++){
                for(j=0;j<r;j++){
                        scanf("%d",&max[i][j]);
                }
        }
        printf("Enter Available Resources:\n");
        for(i=0;i<r;i++)
        {
                scanf("%d",&avail[i]);
        }
        for(i=0;i<n;i++)
        {
                finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        need[i][j]= max[i][j]-alloc[i][j];
                }
        }
        while(count=0)
        {
                safeFound = 0;
                for(i=0;i<n;i++)
                {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                        need[i][j]= max[i][j]-alloc[i][j];
                }
        }
        while(count=0)
        {
                safeFound = 0;
                for(i=0;i<n;i++)
                {
                        if(finish[i] == 0)
                        {
                                int safe =1;
                                for(j=0;j<r;j++)
                                {
                                        if(need[i][j] > avail[j])
                                        {
                                                safe=0;
                                                break;
                                        }
                                }
                                if(safe)
                                {
                                        for(k=0;k<r;k++)
                                        {
                                                avail[k] = avail[k]+alloc[i][k];
                                        }
                                        safe_sequence[count]=1;
                                        finish[i]=1;
                                        safeFound=1;
                                        count++;
                                }
                        }
                }
                if(!safeFound)
                {
                        printf("System is i unsafe state! May occur the deadlock \n");
                        return 0;
                }
        }
        printf("Safe sequenceis:");
        for(i=0;i<n;i++)
        {
                printf("P%d ",safe_sequence[i]);
        }
        return 0;
}
```

```
[rene@localhost ~]$ vi bankers.c
[rene@localhost ~]$ gcc bankers.c
[rene@localhost ~]$ ./a.out
Enter number of processes 3
Enter number of resources types 3
Enter Aloocation Matrix:
1 0 0
1 1 0
1 1 1
Enter Max Matrix:
2 1 1
1 2 1
3 3 2
Enter Available Resources:
1 1 1
Safe sequenceis:P0 P0 P-1320512168 [rene@localhost ~]$
```